

Multiple Choice (10 marks)

1. Occasionally a boy is born with undescended testes (cryptorchidism), and he will be sterile unless they are surgically caused to descend into the scrotum. Explain the reason for this.
 - (a) Human sperm cannot develop at the high temperature found within the body cavity.
 - (b) The testes require the lower pressure environment of the scrotum for sperm maturation
 - (c) Testes must be in the scrotum to properly attach to the vas deferens
 - (d) The scrotum provides the necessary nutrients for sperm development
 - (e) Gravity is required for the testes to function properly
2. Sea divers are aware of a danger known as the "bends". Explain the physiological mechanism of the bends.
 - (a) Carbon dioxide bubbles forming in the blood causing acidosis
 - (b) Rapid increase in barometric pressure causing the bends
 - (c) Formation of helium bubbles in the bloodstream due to rapid ascent
 - (d) Increased salinity in the water leading to dehydration and joint pain
 - (e) Oxygen bubbles forming in the blood leading to hypoxia
3. What is the function of the contractile vacuole?
 - (a) Protects the cell from foreign invaders
 - (b) Acts as a storage for genetic material
 - (c) Regulates the pH level within the cell
 - (d) Transports oxygen throughout the cell
 - (e) Removes nitrogenous wastes
4. Why is genetic drift less important in human evolution at present than it was twenty thousand or more years ago?
 - (a) The human genome has fully adapted to current environmental pressures
 - (b) Modern medicine has eliminated the effects of genetic drift
 - (c) The rate of genetic drift is directly proportional to technological advancements
 - (d) Humans have reached a genetic equilibrium where drift is no longer possible
 - (e) Humans have developed immunity to genetic drift

5. Chromosomal sex in mammals has been shown to be strongly Y chromosome determined. How was this shown to be true?
 - (a) The absence of a Y chromosome determines maleness in mammals
 - (b) The presence of a Y chromosome usually determines maleness in mammals
 - (c) The ratio of two X chromosomes to one Y chromosome determines maleness
 - (d) The presence of an X chromosome determines maleness in mammals
 - (e) The presence of an X chromosome in the absence of a Y chromosome determines maleness

True / False (10 marks)

Answer True or False

6. True or False: The light-dependent reactions produce NADP⁺, ADP, and sugar as products during photosynthesis.
7. True or False: A gal^+ cell will grow rapidly, forming a large colony when cultured in galactose as the carbon source.
7. True or False: Proteins destined for the lysosome require only one signal peptide for their targeting and localization.
8. True or False: The typical periods in the unfolding of a behavioral act include appetitive behavior, consummatory act, and quiescence.
9. True or False: The acrosome and cortical reactions ensure that one and only one sperm will fertilize the egg, preventing the formation of twins.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given a gene for widow's peak with a frequency of 0.07, calculate and discuss the expected frequencies of homozygous and heterozygous individuals for this trait, as well as those lacking it.
12. Discuss the concept of transgressive segregation in genetics, explaining its implications for variability in the F₂ generation and its significance in evolutionary biology.

End of Paper